INCH-POUND

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SUPERSEDING
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PERFORMANCE SPECIFICATION

FUEL CONTROL, MAIN, ELECTROHYDRAULIC

This specification is approved for use by the U.S. Army Tank Automotive Research, Development and Engineering Center (TARDEC), Research, Development and Engineering Command (RDECOM), Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an electrohydraulic main fuel control used in the gas turbine engine of the M1 tank, referred to herein as "assembly." The assembly is an electrohydraulic device which converts electrical commands to fuel flow and power turbine stator (PTS) inlet guide vane (IGV) actuator flows and pressures. This assembly also contains the PTS and IGV electrical feedback components.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

Comments, suggestions, or questions on this document should be addressed to U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-EN/STND/TRANS MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to DAMI_STANDARDIZATION@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

FEDERAL SPECIFICATIONS

A-A-52557 - Fuel Oil, Diesel; For Posts, Camps and Stations

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-461 - Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-810 - Environmental Engineering Considerations and Laboratory Tests

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

U.S. ARMY TACOM DRAWINGS

12324614 - Fuel Control, Main - Electrohydraulic
12286971 - Power Turbine Stator

(Copies of these drawings are available from DAMI_STANDARDIZATION@conus.army.mil or U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-EN/STND/TRANS MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

GENERAL DYNAMICS DEFENSE GROUP

LS199500011841 - Nuclear Hardening Criteria (U)

(Copies of these documents are available from DAMI_STANDARDIZATION@conus.army.mil or U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-EN/STND/TRANS MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000.)

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ASSOCIATION CONNECTING ELECTRONICS INDUSTRIES (IPC)

IPC-TM650.2.6.1 - Fungus Resistant Printed Wiring Materials

(Applications for copies may be obtained from Association Connecting Electronics Industries, 3000 Lakeside Drive, 309 S, Bannockburn, IL 60015 or http://www.ipc.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Materials. Materials shall be as specified herein and in referenced specifications and standards (see 4.5.1).

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs (see 4.5.1).

3.2.2 Surface treatment. Unless otherwise specified (see 6.2), surfaces other than mounting surfaces shall be treated in accordance with the manufacturer's commercial practices and shall meet the requirements specified herein (see 4.5.1).

3.3 Design and construction. The assembly shall be designed and constructed as specified herein, and in accordance with Drawing 12324614 and its applicable standards and specifications (see 4.5.2).

3.3.1 Nuclear hardening. The fuel control assembly shall be designed to withstand the nuclear threat environments as defined in General Dynamics Engineering classified (confidential) document entitled "Nuclear Hardening Criteria (U)", document control number LS199500011841 (see 4.5.1 and 4.5.2).

3.3.2 Interchangeability. The assembly shall be interchangeable on existing equipment without modification or adjustments (see 4.5.2).

3.3.3 Weight. The weight of the assembly shall be no greater than 35 pounds (lb) (see 4.5.2).

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3.4 Performance. Assemblies shall meet the following performance requirements listed below as 3.4.1 through 3.4.11 with their respective tolerances listed in table I. Performance operation shall require diesel fuel conforming to No. 2-D of A-A-52557.

TABLE I. Tolerances.

Function	20 °F .856 sp. gr. 1/	150 °F .800 sp. gr.	85 °F .830 sp. gr.
Minimum W_f *2/	43±5 pph	40±5 pph	42±5 pph
0.60 volts req *	61±6 pph	58±6 pph	60±5 pph
1.50 volts req *	150±12 pph	142±12 pph	147±11 pph
2.00 volts req *	199±13 pph	189±13 pph	195±12 pph
4.00 volts req *	411±25 pph	390±25 pph	403±17 pph
7.00 volts req *	714±28 pph	679±28 pph	700±19 pph
Maximum W_f *	814±30 pph	775±30 pph	799±21 pph
Back-up W_f *	128±11 pph	121±11 pph	125±0 pph
Relief valve	1125±60 psig	1125±60 psig	1125±45 psig
Cut-off valve leak	7.0 cm^3/m max	7.0 cm^3/m max	5.0 cm^3/m max
PTS F/B @ 5.140"	-6.50±0.10 volts	-6.50±0.10 volts	-6.50±0.05 volts
IGV F/B @ 5.470"	-0.67±0.15 volts	-0.67±0.15 volts	-0.67±0.08 volts
PTS actuator 9 ma	20 - 160 psid 3/	20 - 160 psid	30 - 150 psid
PTS actuator 900 ma	20 - 160 psid	20 - 160 psid	30 - 150 psid
IGV actuator 0 ma	20 - 85 psid	20 - 85 psid	30 - 75 psid
IGV actuator 900 ma	20 - 85 psid	20 - 85 psid	30 - 75 psid
Pressure test	1100 psig	1100 psig	1100 psig

1/ Sp. gr. = specific gravity
2/ W_f = fuel flow
3/ PSID = pounds per square inch differential
* = metering valve position
1.0 volt = 100 pph reference fuel @ 60 °F

NOTES:
(1) Reference fuel @ 20 °F is 12 centistokes
(2) Tolerance at 20 °F fluid temperature applies at fuel handling unit (fhu) ambient @ -70 °F.

3.4.1 Minimum fuel flow. The assembly shall provide a fuel flow of no less than 42±2 pounds per hour (pph) (see 4.5.3.1).

3.4.2 Backup valve operation. The assembly shall supply a fuel flow of 118±6 pph with loss of electrical power to the assembly (see 4.5.3.2).

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3.4.3 Maximum fuel flow. The assembly shall supply a fuel flow of 799±16 pph (see 4.5.3.3).

3.4.4 High pressure relief valve operation. The assembly shall provide a pressure relief valve to limit the pressure at the control discharge port to 1150±25 pounds per square inch gage (psig) (see 4.5.3.4).

3.4.5 Fuel flow schedule. The assembly shall schedule fuel flow in accordance with figure 3, with null current set to 275±5 milliamperes (mA) at 0.144 volts root mean square (Vrms) (see 4.5.3.5).

3.4.6 Cutoff valve and check valve operation. The assembly shall provide a combination cutoff and fuel flow check valve. The cutoff valve shall be normally open and energized closed with leakage not to exceed 5 cubic centimeter per minute (cm^3/min) at 920±10 psig. The check valve shall open at 54±5 psig with leakage not to exceed 0.25 cm^3/min at 40 to 42 psig (see 4.5.3.6).

3.4.7 Power turbine stator (PTS) feedback voltage output. The assembly shall provide an electrical feedback signal proportional to the positioning of the PTS feedback lever. The position shall be 5.135 to 5.145 inches (in.) in accordance with 12286971. The relationship of the PTS secondary voltage versus this dimension, for an excitation voltage of 5.773 Vrms on the primary, shall be as follows:

Specified dimension	PTS secondary voltage
5.140±0.001 in.	1.2499±0.0192 Vrms
6.122±0.001 in.	0.1192±0.0577 Vrms

For specified pin definition, see figure 1 (see 4.5.3.7).

3.4.8 Inlet guide vane (IGV) feedback voltage output. The assembly shall provide an electrical feedback signal proportional to the positioning of the IGV feedback lever. The position shall be 5.465 to 5.475 in. in accordance with Drawing 12286971. The relationship of the IGV secondary voltage versus this dimension for an excitation voltage of 5.773 Vrms on the primary shall be as follows:

Specified dimension	IGV secondary voltage
5.470±0.001 in.	0.0769±0.0115 Vrms
7.040±0.001 in.	1.2302±0.0346 Vrms

For specified pin definition, see figure 1 (see 4.5.3.8).

3.4.9 PTS actuator flow. The assembly shall provide a relative flow between ports 7 and 8, in accordance with Drawing 12286971, as a function of electrical current to the PTS solenoid. The relationship of electrical current to the flow shall be as follows (see 4.5.3.9):

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PTS solenoid current	Pressure/flow requirements
(Pins g and h)	350 psid P_1 - P_0 across the fhu.
0 (+10/-0) ma	When port 7 is 24 to 150 psid greater than port 8, actuator flow shall be 220 to 570 pph.
350±50 ma	When port 7 is equal to port 8, actuator flow shall be 0 pph and pressure shall be equal.
900±5 ma	When port 8 is 24 to 150 psid greater than port 7, actuator flow shall be 220 to 570 pph.

3.4.10 IGV actuator flow. The assembly shall provide a relative flow between ports 5 and 6 as a function of electrical current to the IGV solenoid. The relationship of electrical current to the flow shall be as follows (see 4.5.3.10):

IGV solenoid current	Pressure/flow requirements
(Pins a and b)	350 psid P_1 - P_0 across the fhu.
0 (+10, -0) ma	When port 6 is 15 to 75 psid greater than port 5, actuator flow shall be 90 to 210 pph.
350±50 ma	When port 6 is equal to port 5, actuator flow shall be 0 pph and pressure shall be equal.
900±5 ma	When port 5 is 15 to 75 psid greater than port 6, actuator flow shall be 90 to 210 pph.

3.4.11 Pressure resistance. The assembly shall withstand a pressure test of 1100 psig with no leakage (see 4.5.3.11).

3.5 Environmental. The assembly shall meet the operational performance requirements specified in 3.4 during and after, as applicable, the following environmental conditions of 3.5.1 through 3.5.9.

3.5.1 High temperature. The assembly shall operate as specified in 3.4 after exposure to an ambient temperature of 250±10 °F, with an inlet fuel temperature of 150±10 °F (see 4.5.4.1).

3.5.2 Low temperature. The assembly shall operate as specified in 3.4 and achieve 10 simulated starts during and after exposure to an ambient temperature of -60±10 °F with fuel viscosity ranging from 4.1 to 5.3 centistokes (see 4.5.4.2).

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3.5.3 Gun firing shock resistance. The assembly shall operate as specified in 3.4 during and after exposure to shocks of 55±5.5 g for 2.5±0.2 milliseconds (ms) duration half sine wave in each direction of three mutually perpendicular axes (see 4.5.4.3).

3.5.4 Ballistic shock resistance. The assembly shall operate as specified in 3.4 after subjection to shocks of 200±20 gravity units (g) for 0.5±0.1 ms duration half sine wave in each direction of three mutually perpendicular axes (see 4.5.4.4).

3.5.5 Basic shock resistance. The assembly shall operate as specified in 3.4 during and after exposure to shocks of 20±2 g for 18 ms duration half sine wave in each direction of three mutually perpendicular axes (see 4.5.4.5).

3.5.6 Vibration resistance. The assembly shall operate as specified in 3.4 during and after subjection to the peak vibration envelope of figure 2 in each of three mutually perpendicular axes. The vibration shall be imposed at a logarithmic sweep rate of 20 minutes per sweep cycle from 5 to 500 to 5 Hertz (Hz) (see 4.5.4.6).

3.5.7 Submergence. The assembly shall operate as specified in 3.4.9 and 3.4.10 to regulate the required fuel flow rate and the flow and pressure to the PTS (3.4.9) and IGV (3.4.10) servo valves when subjected to a water submergence of 6 feet (ft) (see 4.5.4.7).

3.5.8 Sand and dust. The assembly shall operate as specified in 3.4 during and after exposure to a sand and dust environment (see 4.5.4.8).

3.5.9 Fungus. The assembly shall operate as specified in 3.4 after exposure to conditions favorable to fungus growth. Unless otherwise specified (see 6.2), these conditions shall be as defined in IPC TM-650 2.6.1 (see 4.5.4.9).

3.5.10 Electromagnetic interference and compatibility. The assembly shall be capable of meeting the emission and susceptibility requirements of MIL-STD-461, Tests CE101, CE102, CE106, CS101, CS114, CS116, RE102 and RS103 (see 4.5.4.10).

3.6 Endurance and reliability.

3.6.1 Endurance. The endurance of the assembly shall be based on satisfying the performance requirements of 3.4 after 350 hours of operation (see 4.5.4.11 and 6.2) in accordance with the fuel flow schedule of figure 3 and the duty cycle of figure 4.

3.6.2 Reliability. The reliability of the assembly shall be demonstrated based on achieving the mean time between failures in hours (MHBF) listed below upon completion of endurance operations (see 4.5.4.11 and 6.2). The MHBFs shall be calculated by dividing the total endurance operating hours by the number of reliability failures defined below:

a. 10 000 MHBF for failures causing loss of total fuel schedule

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b. 4800 MHBF for failures causing fuel restriction resulting in a major fuel schedule change, defined as a 20% loss of fuel flow [564 - 600 pph] at rated power
c. 1200 MHBF for failures causing unscheduled maintenance action

3.7 Identification and marking. The identification and marking shall include, as a minimum, the department or agency identification, the registration number, and legend as specified for the appropriate military agency (see 6.2). Marking shall be permanent and legible. When specified (see 6.2), concealed marking shall be furnished (see 4.5.2).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
    1. Preproduction inspection (see 4.3.1)
    2. Initial production inspection (see 4.3.2)
b. Conformance inspections (see 4.4)
    1. Examination (see 4.4.2)
    2. Tests (see 4.4.3)

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the environmental test conditions as follows:

a. Fuel temperature: 85±5 °F
b. Air temperature: 73±18 °F
c. Barometric pressure: 25.5 to 30.5 in. of mercury (Hg)
d. Relative humidity: 50±30%

4.2.1 Test equipment. The following test equipment shall be used in the performance of the examinations and test specified herein:

a. Test set, Bendix model EK-31A or equivalent
b. Fixture assembly, Bendix part number 2524777-G1 or equivalent

4.3 First article inspection. First article inspection shall be performed on preproduction or initial production samples as specified herein. Approval of the first article sample by the Government shall not relieve the contractor of the obligation to supply assemblies that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.3.1 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of two assemblies. Preproduction inspection shall consist of the inspections in the order listed and as specified in table II, except testing for endurance and reliability. When

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specified (see 6.2), endurance and reliability test will be conducted on preproduction samples, as specified in 6.2. Samples undergoing endurance and reliability testing shall not be subjected to environmental tests of 4.5.4.1 through 4.5.4.9.

4.3.2 Initial production inspection. Unless otherwise specified (see 6.2), the Government shall select two assemblies of the first ten assemblies produced under the production contract for initial production inspection. Initial production units shall be inspected in the order listed and as specified in table II.

TABLE II. Classification of inspections.

Title	Requirement	Inspection	First article	Conformance	
				Examination	Tests
Materials and construction	3.2, 3.2.1, 3.2.2 and 3.3.1	4.5.1	X	X	X
Defects (see 4.5.2 and table III)	3.3 thru 3.3.3, and 3.7	4.5.2	X	X	
Minimum fuel flow	3.4.1	4.5.3.1	X		X
Backup valve operation	3.4.2	4.5.3.2	X		X
Maximum fuel flow	3.4.3	4.5.3.3	X		X
High pressure relieve valve operation	3.4.4	4.5.3.4	X		X
Fuel flow schedule	3.4.5	4.5.3.5	X		X
Cutoff valve and check valve operation	3.4.6	4.5.3.6	X		X
PTS feedback voltage output	3.4.7	4.5.3.7	X		X
IGV feedback voltage output	3.4.8	4.5.3.8	X		X
PTS actuator flow	3.4.9	4.5.3.9	X		X
IGV actuator flow	3.4.10	4.5.3.10	X		X
Pressure resistance	3.4.11	4.5.3.11	X		X
High temperature	3.5.1	4.5.4.1	X		
Low temperature	3.5.2	4.5.4.2	X		
Gun firing shock resistance	3.5.3	4.5.4.3	X		
Ballistic shock resistance	3.5.4	4.5.4.4	X 1/		
Basic shock resistance	3.5.5	4.5.4.5	X		
Vibration resistance	3.5.6	4.5.4.6	X		
Submergence	3.5.7	4.5.4.7	X		
Sand and dust	3.5.8	4.5.4.8	X		
Fungus	3.5.9	4.5.4.9	X		
Electromagnetic interference and compatibility	3.5.10	4.5.4.10	X 1/		
Endurance	3.6.1	4.5.4.11.1	X 2/		
Reliability	3.6.2	4.5.4.11.2	X 2/		

1/ Preproduction only.
2/ Samples shall not be subjected to tests 4.5.4.1 through 4.5.4.9.

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4.4 Conformance inspections.

4.4.1 Sampling. Sampling shall be as specified in the contract or order (see 6.2).

4.4.1.1 Lot formation. An inspection lot shall consist of all the assemblies of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.4.2 Examination.

4.4.2.1 Classification of defects. For examination purposes, defects shall be classified as listed in table III.

TABLE III. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major:		
101	Dimensions affecting interchangeability, not within tolerance (see 3.3 and 3.3.2).	SIE 1/
102	Improper weight (see 3.3.3).	Scale
103	Faulty workmanship affecting performance (see 3.3.1).	Visual
Minor:		
201	Dimensions not affecting interchangeability, not within tolerance (see 3.3).	SIE
202	Improper marking (see 3.7).	Visual

1/ SIE = Standard Inspection Equipment.

4.4.3 Acceptance tests (100 percent). Each assembly shall be subjected to the tests specified in table II, and shall be free from defects.

4.5 Methods of inspection.

4.5.1 Materials. Conformance to 3.2 through 3.2.2, and 3.3.1 shall be determined by inspection of contractor records providing proof or certification that materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.5.2 Defects. Conformance to 3.3 through 3.3.3, and 3.7 shall be determined by examination for the defects listed in table II. Examination shall be visual, tactile, or by measurement with SIE.

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4.5.3 Performance tests.

4.5.3.1 Minimum fuel flow. To determine conformance to 3.4.1, the fuel control assembly shall be tested using test equipment specified in 4.2.1 and military aircraft type II calibrating fluid (see 6.3) or commercial equivalent.

4.5.3.2 Backup valve operation. To determine conformance to 3.4.2, electrical power to the assembly shall be off and the backup fuel flow shall be measured using the test equipment specified in 4.2.1 and military aircraft type II calibrating fluid (see 6.3) or commercial equivalent.

4.5.3.3 Maximum fuel flow. To determine conformance to 3.4.3, the fuel flow shall be measured using test equipment specified in 4.2.1 and military aircraft type II calibrating fluid (see 6.3) or commercial equivalent.

4.5.3.4 High pressure relief valve. To determine conformance to 3.4.4, the assembly shall be visually examined to verify the provision of a relief valve. The assembly shall be tested using equipment specified in 4.2.1 and military aircraft type II calibrating fluid (see 6.3) or commercial equivalent to measure pressure at the control discharge port.

4.5.3.5 Fuel flow schedule. To determine conformance to 3.4.5, the assembly scheduled fuel flow shall be measured using test equipment specified in 4.2.1 and military aircraft type II calibrating fluid (see 6.3) or commercial equivalent.

4.5.3.6 Cutoff valve and check valve operation. To determine conformance to 3.4.6, the assembly shall be examined to verify the provision of a combination cutoff and fuel flow check valve. The assembly shall be tested for opening, closing and leakage with equipment specified in 4.2.1 and type II calibrating fluid (see 6.3) or commercial equivalent.

4.5.3.7 PTS feedback voltage output. To determine conformance to 3.4.7, the position, specified dimension and voltage shall be measured using test equipment specified in 4.2.1 and military aircraft type II calibrating fluid (see 6.3) or commercial equivalent.

4.5.3.8 IGV feedback voltage output. To determine conformance to 3.4.8, the position, specified dimension and voltage shall be measured using test equipment specified in 4.2.1 and military aircraft type II calibrating fluid (see 6.3) or commercial equivalent.

4.5.3.9 PTS actuator flow. To determine conformance to 3.4.9, the relationship of the electrical current to the flow shall be measured using test equipment specified in 4.2.1 and military aircraft type II calibrating fluid (see 6.3) or commercial equivalent.

4.5.3.10 IGV actuator flow. To determine conformance to 3.4.10, the relationship of the electrical current to the flow shall be measured using test equipment specified in 4.2.1 and military aircraft type II calibrating fluid (see 6.3) or commercial equivalent

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4.5.3.11 Pressure resistance. To determine conformance to 3.4.1l, the assembly shall be tested to withstand 1100 psig with no leakage, using test equipment specified in 4.2.1 and military aircraft type II calibrating fluid (see 6.3) or commercial equivalent.

4.5.4 Environmental. Assemblies undergoing first article inspections shall meet the operational performance requirements of 3.4 during and after, as applicable, the environment testing specified in 4.5.4.1 through 4.5.4.9. Recalibration shall not be performed during or after subjection to environmental testing.

4.5.4.1 High temperature. To determine conformance to 3.5.1, the assembly shall be mounted in a temperature chamber in operational condition on a base whose temperature shall be maintained at 300±20 °F. Ambient room temperature shall be maintained at 250±10 °F and the No. 2-D (A-A-52557) fuel shall be maintained at an inlet temperature of 150±10 °F. The fuel shall be contaminated with water to a level of 0.01% by volume.

a. The assembly shall be subjected to 8 hours of duty cycle, as defined in figure 4, while maintained at the above specified conditions. The assembly shall meet the performance requirements of 3.4 while operating under these conditions.
b. The assembly shall be returned to standard ambient conditions specified in 4.2, nonoperating, and stabilized.
c. The assembly shall then be operated under standard ambient conditions and meet performance requirements specified in 3.4.

4.5.4.2 Low temperature. To determine conformance to 3.5.2, the assembly shall be mounted in a temperature chamber in operational condition. The ambient air temperature shall be -60±10 °F with fuel viscosity ranging from 4.1 to 5.3 centistokes.

a. The assembly shall temperature soak for 8 hours at the conditions specified above. The assembly shall achieve 10 simulated starts under these conditions.
b. The assembly shall be returned to standard ambient conditions specified in 4.2 and stabilized. The assembly shall then be operated under these conditions and shall again achieve 10 simulated starts.

4.5.4.3 Gun firing shock resistance. To determine conformance to 3.5.3, the assembly shall be mounted, using the installed interface configuration, and in operational condition. A total of three half sine wave shock pulses shall be applied in both directions along each of three mutually perpendicular axes (total of 18 shocks). Peak amplitude shall be 55±5.5 g for duration of 2.5±0.2 ms. During shock exposure the assembly shall be operating and shall be monitored to verify conformance to the operating performance requirements of 3.4. Upon completion of the shock test, the assembly shall again be inspected to meet the operational performance of 3.4.

4.5.4.4 Ballistic shock resistance. To determine conformance to 3.5.4, the assembly shall be mounted using the installed interface configuration in a nonoperational and pressurized mode. The assembly shall be subjected to three half sine wave shocks of 200±20 g for a duration

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of 0.5±0.1 ms in each direction along each of three mutually perpendicular axes (total of 18 shocks). Upon completion of the shock test, the assembly shall be inspected to meet the operational performance requirements of 3.4.

4.5.4.5 Basic shock resistance. To determine conformance to 3.5.5, the assembly shall be mounted in operational condition using the installed interface configuration. A total of three half sine wave shock pulses of 20±2 g shall be applied in both directions along each of three mutually perpendicular axes for a duration of 18 ms. During exposure, the assembly shall be operating and shall be monitored to verify conformance to the operating performance requirements of 3.4. Upon completion of the shock test, the assembly shall again be inspected to meet the operational performance of 3.4.

4.5.4.6 Vibration resistance. To determine conformance to 3.5.6, the assembly shall be mounted in operational condition using the installed interface configuration. Sinusoidal vibration tests shall be conducted over a period of 6 hours. During this test the assembly shall be operating and the following test conditions shall be applied:

a. The vibration curve shall be in accordance with figure 2.
b. The ambient room temperature shall be 140 °F.
c. The frequency of applied vibration shall be 5 to 500 to 5 Hz within a logarithmic resonant sweep time of 20 minutes. The sweep shall be conducted in each of the three mutually perpendicular axes, to identify up to four of the most severe resonant frequencies. A resonant dwell shall then be conducted for 20 minutes at each of the four resonant frequencies, for a total of 120 minutes in each axis. If less than four resonant frequencies are found, the balance of the 120 minute vibration time in each axis shall be spent cycling the assembly over the specified frequency range.
d. During exposure, the assembly shall be operating and shall be monitored to verify conformance to the operating performance requirements of 3.4. Upon completion of the vibration test, the assembly shall again be inspected to meet the operational performance requirements of 3.4.

4.5.4.7 Submergence. To determine conformance to 3.5.7, all fluid and electrical fittings shall be capped and the assembly shall be subjected to a one-hour cycle test consisting of immersion in 70 °F water from a depth of 6 ft to 8 ft at a rate of 4 cycles per minute. Within two hours after completion, the calibration shall be checked to verify conformance of fuel flow rate and the flow and pressure to the PTS (3.4.9) and IGV (3.4.10) servo valves.

4.5.4.8 Sand and dust. To determine conformance to 3.5.8, the fuel control assembly (with the PTS and IGV feedback assemblies installed and rigged for cycling stop to stop) shall be subjected to a sand and dust environment for a period of 10 hours. All other body portions and openings shall be capped. Prior to test, 5 ounces of military petroleum base (see 6.3) or commercial equivalent hydraulic fluid shall be poured over each feedback mechanism. Both lever assemblies shall be rigged to cycle from full open to full closed every 30 seconds, with a 2 degree dither cycle at a rate of 2 cycles per second superimposed for a 10-second interval. A stream of dust-laden air 0.1 to 0.5 gram per cubic foot (g/ft^3) density shall be directed at the

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feedback assemblies at a velocity of 600 to 1200 feet per minute (ft/min). Test chamber temperature and relative humidity of 80±10 °F and 10 percent maximum, respectively, shall be maintained during the test of 10 hours duration. Lever running torques and static feedback voltage readings of several positions shall be recorded before, during, and after tests.

4.5.4.9 Fungus. To determine conformance to 3.5.9, the fuel control assembly shall be tested in accordance with MIL-STD-810. In lieu of fungus testing, certification (see 4.5.1) with supporting data may be provided attesting that the assembly is constructed of materials that would not support fungus growth.

4.5.4.10 Electromagnetic interference and compatibility. To determine conformance to 3.5.10, the fuel control assembly shall be tested in accordance with MIL-STD-461.

4.5.4.11 Endurance and reliability. To demonstrate conformance to 3.6.1 and 3.6.2, the assembly shall be operated using the test equipment specified in 4.2.1 or equivalent, and in accordance with the fuel flow schedule and duty cycle specified in 3.6.1.

4.5.4.11.1 Endurance test. To demonstrate conformance of 3.6.1, the assembly shall be operated for a total of not less than 350 hours. During testing, reliability failures (see 3.6.2) and other deficiencies shall be recorded. Repair/replacement of failed items shall be made in order to complete 361.4 hours of operation. Upon completion of the endurance operating hours, the performance tests of 3.4 shall be conducted.

4.5.4.11.2 Reliability verification. To determine conformance to 3.6.2, the reliability of the assembly, in MHBF, shall be calculated by multiplying the endurance operating hours by the number of samples and dividing the result by the number of reliability failures recorded under 4.5.4.11.1

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

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6.1 Intended use. The fuel control assembly is to be used in the fuel management system on the AGT 1500 turbine engine.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. If required, the specific issue of individual documents referenced (see 2.2.1, 2.2.2, and 2.3).
c. If first article inspection is not required (see 3.1).
d. If surface treatments are other than as specified (see 3.2.2).
e. If conditions for fungus growth are other than as specified (see 3.5.9).
f. If endurance and reliability testing is required (see 3.6.1, 3.6.2 and 4.3.1).
g. Sample size for endurance and reliability testing (see 3.6.1, 3.6.2 and 4.3.1).
h. Identification of appropriate service for marking (see 3.7).
i. Concealed markings, if required (see 3.7).
j. If inspection conditions are other than as specified (see 4.2).
k. If preproduction samples are required (see 4.3.1).
l. If preproduction samples are other than as specified (see 4.3.2).
m. Sampling if required (see 4.4.1).
n. Packaging requirements (see 5.1).

6.3 Military fluids. The following military fluids may be used for testing:

MIL-PRF-6083 - Hydraulic Fluid, Petroleum Base, Preservation and Operation
MIL-PRF-7024 - Calibrating Fluid, Aircraft Fuel System Components

6.4 Subject term (key word) listing.

Check valve
Cutoff valve
Inlet guide ramp
Power turbine starter

6.5 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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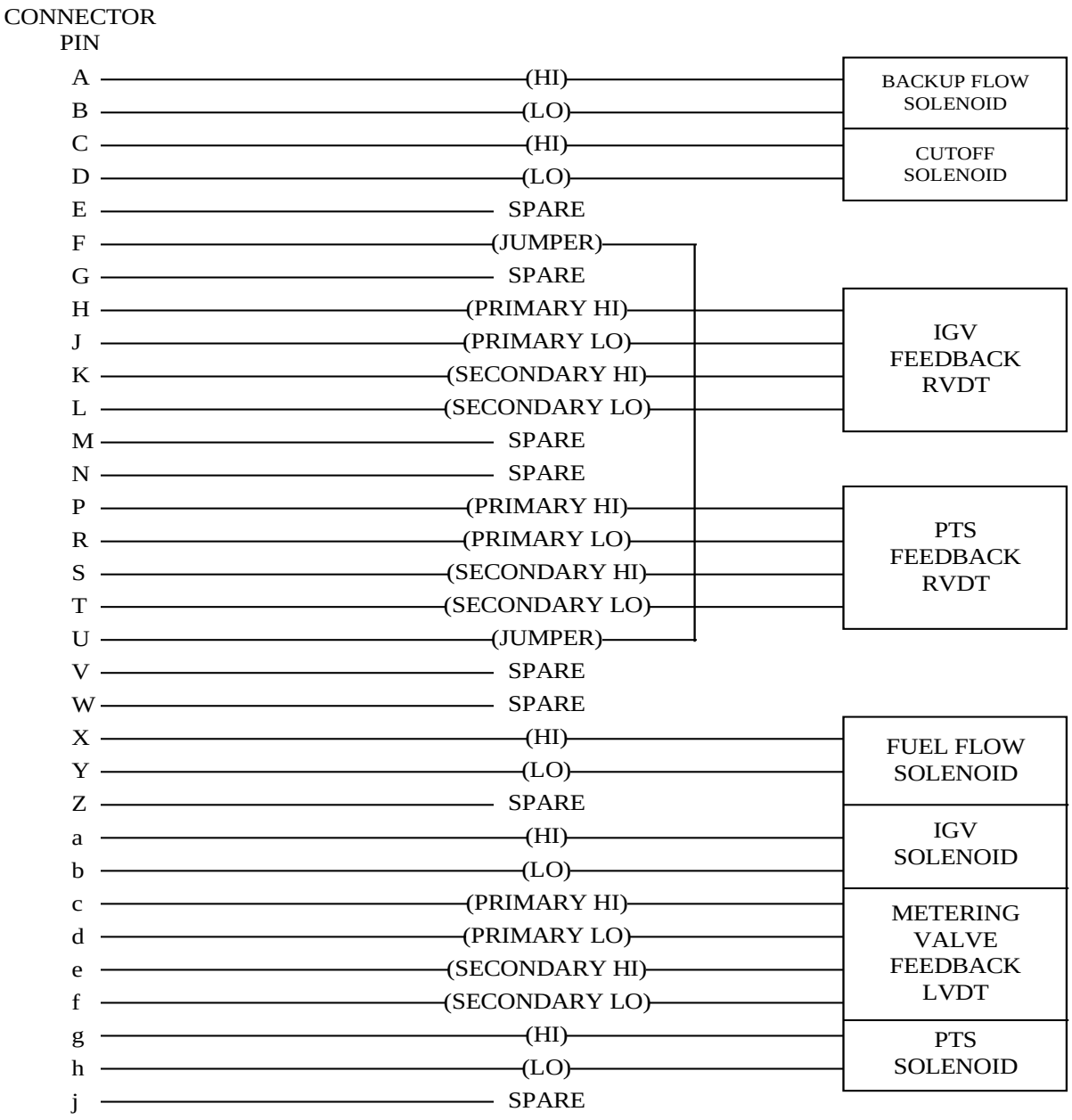


FIGURE 1. Connector pin definition.

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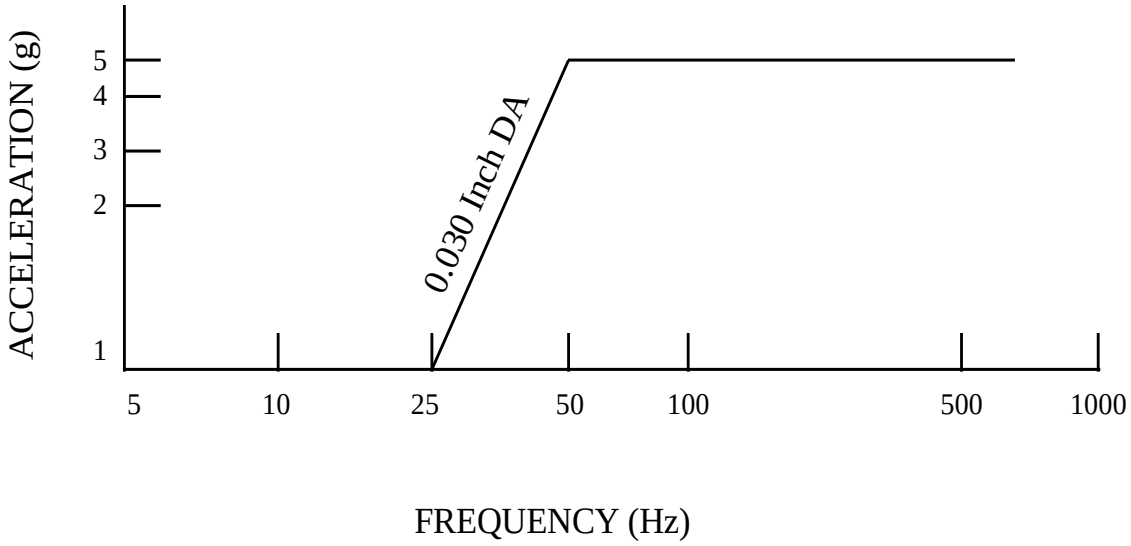


FIGURE 2. Peak vibration envelope.

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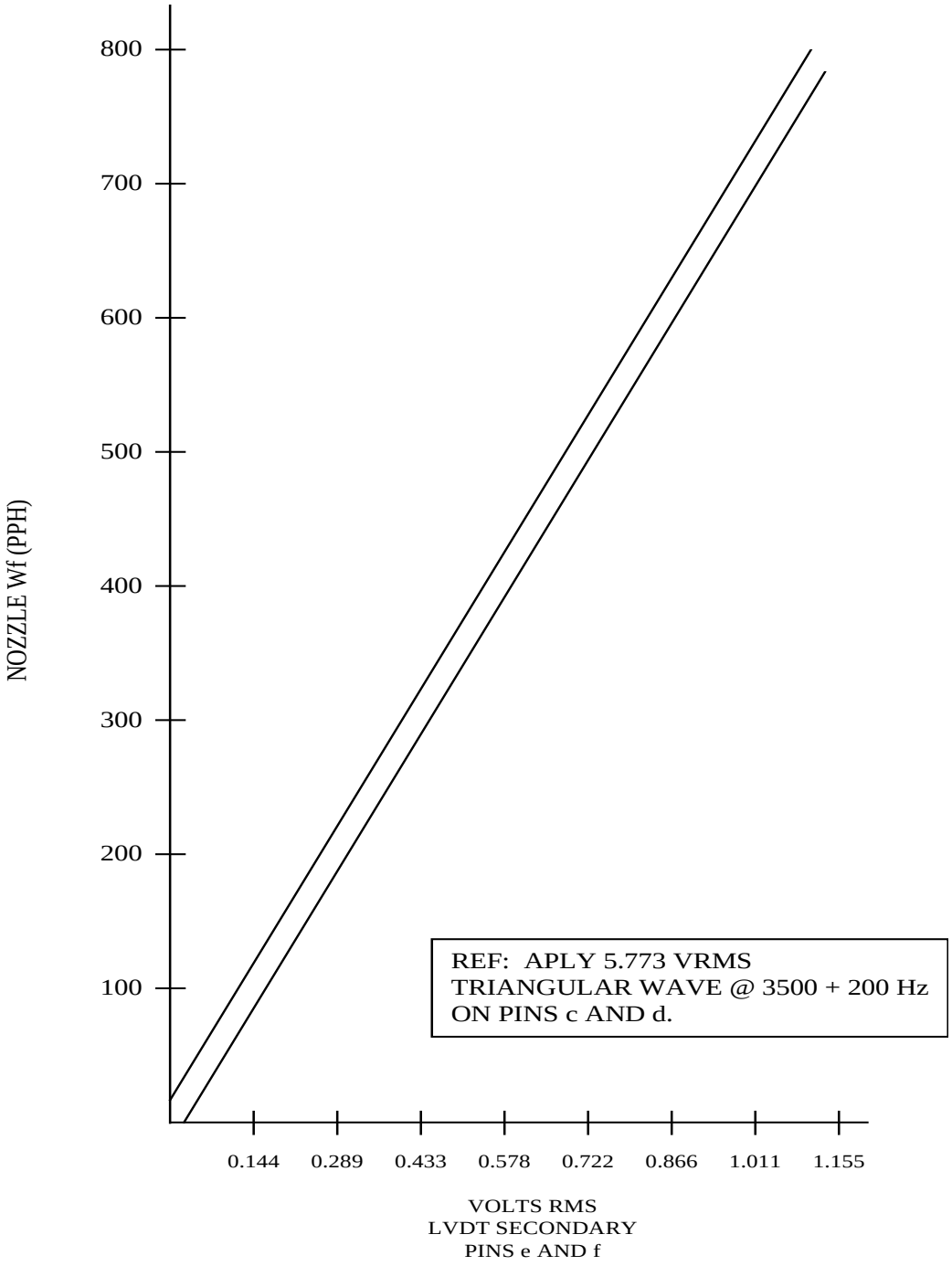


FIGURE 3. Fuel flow schedule.

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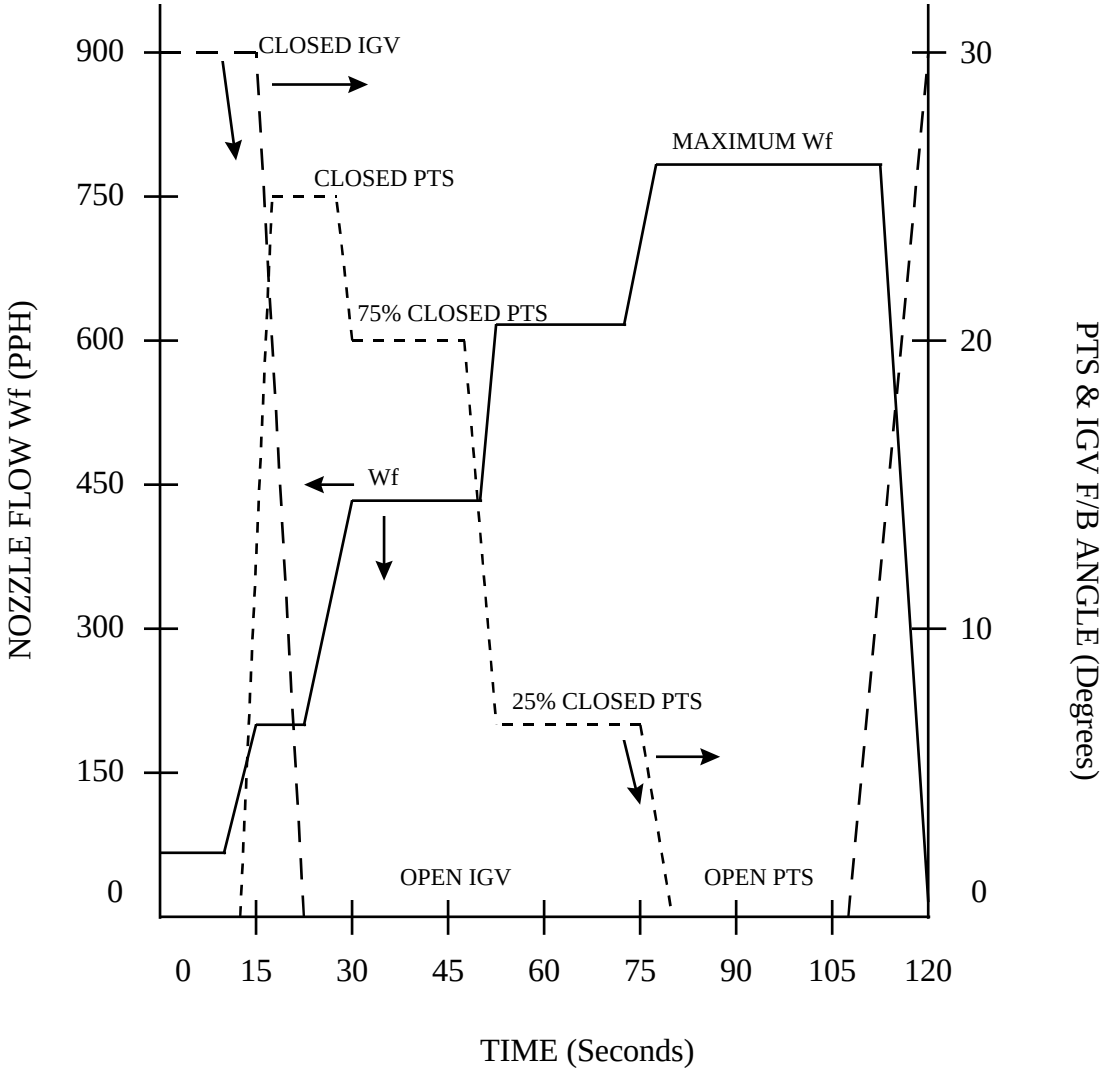


FIGURE 4. Duty cycle.

Custodian:
Army - AT

Preparing Activity:
Army - AT

(Project 2910-2009-006)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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